

两种固体氧化物燃料电池系统的故障建模与仿真对比研究

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摘要: 固体氧化物燃料电池(SOFC)系统的建模方式较多, 而基于机理模型的故障诊断是能够通过系统的动态趋势辨别故障的有效手段之一, 但该方法对机理模型的准确性有要求. 此外, 不同的燃料供给系统采用的系统结构也是有差异的, 进而导致在相同故障下SOFC系统的动态响应也是不同的. 因此, 本文基于两种燃料供应方式, 提出了分别以纯氢气和天然气作为燃料的SOFC系统结构, 并基于原有机理知识进行MATLAB/Simulink系统建模. 经与真实SOFC系统实验对比, 搭建的系统模型能够有效模拟系统在无故障状态下的动态变化; 另外, 在无故障模型的基础上, 分别加入两类常见故障, 其一为风机故障, 其二为燃料供应管路泄露. 最后通过仿真分析, 明确了所搭建模型的合理性, 且发现了两种燃料供应对SOFC系统热响应特性是不同的, 对SOFC系统故障的检测和设备选型具有重要意义.

关键词: 建模仿真; 固体氧化物燃料电池; 故障分析; 纯氢气和天然气系统; 故障仿真; 辅助设备故障

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Fault modeling and simulation comparative study of two solid oxide fuel cell system

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Abstract: There are many modeling methods for solid oxide fuel cell (SOFC) systems. One of the fault diagnosis methods is based on accurate system modeling. In addition, different fuel supply systems adopt different system structures, which lead to different dynamic response of SOFC system under the same fault condition. Therefore, based on two kinds of fuel supply modes, this paper proposes the SOFC system structure using pure hydrogen and natural gas as fuel, respectively. Based on the original knowledge, the system model is built by MATLAB/Simulink. Compared with the actual SOFC system experiment, the system model can simulate effectively the system dynamic under the fault-free condition. Moreover, fuel supply manifold leakage fault and blower fault are added to fault-free model respectively. Finally, through simulation analysis, the rationality of the model is clarified. It is found that the thermal response characteristics of two kinds of fuel supplied SOFC system are different, which has important significance for fault detection and type selection of SOFC system.

Key words: simulation modeling; solid oxide fuel cells (SOFC); failure analysis; pure hydrogen and natural gas system; fault simulation; balance of plant fault

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1 Introduction

Solid oxide fuel cell (SOFC) system is a high temperature power generation equipment which work tem-

perature is 600~800°C and the catalyst is not a noble metal. Compared with other power generation technology, fuel cell power generation system is a higher ef-

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iciency, clean and pollution-free technology without burning and mechanical transmission. Moreover the exhaust heat can be used cyclically^[1-4]. Based on these advantages, the SOFC system has great potential for development^[5-6]. With the requirement increasing of system performance, the structure of SOFC systems is more and more complex. The requirements of maintenance and fault diagnosis for systems are also paid more and more attention^[7]. However, the high temperature and airtight operating environment of the SOFC system makes the difficulty of fault detection increase. Therefore, the development of a real-time fault detection method for SOFC systems has great significance for maintaining the system long-term operation.

In general, the faults of the actual system can be classified based on the difference of the occurrence position and the influence of the system performance^[8-9]. It is usually divided into sensor fault, controller fault, s-tack fault, actuator fault and system level fault. In these fault, system level faults are most common. The relationship between the kind of fault and the reason is often not one to one correspondence. In other words, the occurrence of a fault may be caused by a variety of reasons^[10-12]. The common condition is that the components of the system are invalid such as the abnormal physical characteristics of the device, the flawed design software or some aged components.

At present, many literatures of SOFC systems focus on the application of energy management and hybrid energy control in China^[13-15], and few work focus on fault modeling and diagnosis. To the best knowledge of the author, only the literature^[16] used model predictive control to the SOFC system for fault-tolerant control, and the literature^[17] focused on the life prediction for the system, and so on^[7]. Nevertheless, their fault diagnosis is based on an idealized model. And there is no actual SOFC system to support the fault. In addition, these works are the data processing algorithms research such as neural networks, the SOFC system only is a case to illustrate algorithms. The actual system fault is not highlight in their literatures^[9, 18-19].

In the aspect of modeling, there is no literature to validate the mechanism model of SOFC system. Many SOFC system literatures only focus on the stack model verify rather than the validation of the integrated SOFC system model^[20-21]. Therefore, the model validation in this paper will be aimed at the whole system after integration. At the same time, there is no literature to compare and analyze the thermal response of different fuel supply in SOFC system so as to fault detection. Nevertheless, this paper will study the two systems to help the fault detection and system BOP component selection. The main innovations of this paper are as follows:

1) The rationality of the SOFC system model is verified by experimental data from the SOFC system,

which is more meaningful than the SOFC system literature that only verifies the correctness of the stack model;

2) The thermal response characteristics differences of the two fuel SOFC systems are verified by the actual experiment and the mechanism model of SOFC system;

3) The dynamic response trends of SOFC system under normal and fault conditions were compared. It provides guidance for fault detection of SOFC system and selection of BOP components in the future;

4) The simulated faults are from the actual system rather than the hypothetical system faults, which make the proposed faults more representative and research value.

The rest of the paper is organized as follows. The working principle of two SOFC system and the fault mode are analyzed with combining the system structure in Section 2. The SOFC normal condition and two fault conditions of pure hydrogen and natural gas system are modeled in Section 3. The simulation and result is described with actual SOFC system normal/fault state in Section 4. Two faults are analyzed and discussed in Section 5. Last Section gives the conclusions.

2 SOFC system work principle and fault mode

An integrity SOFC power generation system must be equipped with stack and the auxiliary equipment-balance of plant (i.e. BOP). The common structure is composed of five equipments: high-pressure hydrogen/natural gas tank with matched manifold, blower with matched manifold, heat exchanger, SOFC stack, burner^[22-23]. In addition, SOFC stack power generation must make high temperature fuel and air flow into s-tack so that the stack can discharge by electrochemical reaction^[24].

SOFC system is a complex and coupled system, which includes a fuel supply module, air supply module, heat exchange module, power generate module and exhaust gas combust module^[25-26]. The module is distinguished based on the component function, so as to ensure the independence of the module and a clear fault identify. In addition, the fault may still occur at the sensor or controller, which is ignored in this paper. The structure of pure hydrogen SOFC system and natural gas SOFC system are shown in Fig. 1 and Fig. 2, respectively.

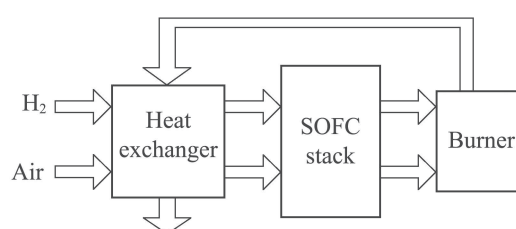


Fig. 1 Structure of pure hydrogen SOFC system

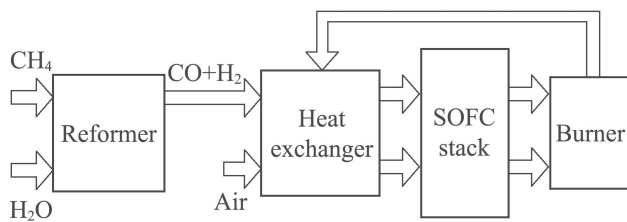


Fig. 2 Structure of natural gas SOFC system

According to this classification, the fault phe-

nomenon of each module can be analyzed. In other words, the fault can be judged based on every abnormal phenomenon. Since the burner technology is mature, the burner fault is ignored in this paper. The common module faults are concentrated in the supply of gas and the changes of stack performance, so the fault can be divided into: anode gas supply (Fuel supply), cathode gas supply (Air supply), heat exchanger and generate electricity. The relative fault classification information can be seen at Table 1.

Table 1 Fault pattern classification

Module	Fault form	Fault result
Fuel (H ₂) supply	Impure fuel	Anodic poisoning, reduced power generation
	H ₂ manifold leakage	Stack temperature decrease down, reduced power generation
Air supply	Air manifold leakage	Stack temperature increase, increased power generation
	Air blower fault	Stack temperature increase, increased power generation
	Air manifold leakage	Stack temperature increase, increased power generation
Heat exchanger	H ₂ manifold leakage	Stack temperature decrease down, reduced power generation
	Exhaust air manifold leakage	Stack temperature decrease down, reduced power generation
Generate electricity	Insufficient fuel supply	Fuel starvation, Anode degradation, reduced power generation
	Insufficient air supply	Stack temperature increase, increased power generation

3 System and fault modeling

In order to obtain a model that express accurately the dynamic response of SOFC system, MATLAB/Simulink 2016 is selected as the modeling software. There are obvious differences between the numerical simulation model focusing on mechanical physical analysis and the mechanism model of matlab. The modeling principle is that the selection of nodes and branches should be helpful to reveal the fault and fault propagation path in an all-round way as far as possible. The modeling steps are as follow:

Step 1 The SOFC system process is divided into several system level units. The process is decomposed into several system level units. The idea of modularization is adopted to decompose the complex process into several simple units at all levels. The modeling of system-level units can greatly simplify the problem.

Step 2 Define variable definitions for each system unit. This step identifies the important variables associated with the system elements, i.e. the nodes of the model for the system elements of the chemical process, which mainly include temperature, flow, and material composition. For specific system units, choose among these five variables.

Step 3 The same variables for each system unit are connected in the direction of mass transfer, according to the Fig. 1 and Fig. 2.

According to the above working principle of each module, we first establish the normal model based on

the physical mechanism^[27–28]. After confirming the correctness of the model, the actuator faults from actual system then are added to the normal model. And the fault function module is simulated. At last, the fault modeling and simulation of the whole SOFC system are obtained.

3.1 Fault-free modeling

Since it is very difficult to modeling the SOFC system accurately, and in order to facilitate the qualitative analysis of the common SOFC system fault, the main system components must be modeled and some assumptions should be used. Assumptions are as follows:

- 1) The system is insulated from the outside;
- 2) The gas is the ideal gas;
- 3) Temperature and pressure are all along the manifold linear changes;
- 4) The stack model is co-directional;
- 5) Supplied gas is distributed uniformly to each channel.

3.2 Stack model

I Quality conservation model.

For the air airway and fuel airway, the mole fraction of the gas component and flow calculation method are the same, that is, when the gas through the node of the SOFC stack, the relationship between the outlet gas flow and the inlet gas flow is^[29]

$$\dot{N}_{\text{out}} = \dot{N}_{\text{in}} + \sum n_i, \quad i \in \{\text{H}_2, \text{O}_2, \text{N}_2, \text{H}_2\text{O}\}, \quad (1)$$

where $\dot{N}_{\text{out}}$ denotes outlet gas flow; $\dot{N}_{\text{in}}$ is inlet gas

flow; n_i is the amount of gas involved in the reaction. The mole fraction of each node is calculated as follows^[29]:

$$N \frac{dX_i}{dt} = \dot{N}_{in} \cdot X_{i,in} - \dot{N}_{out} \cdot X_{i,out} + n_i, \quad i \in \{H_2, O_2, N_2, H_2O\}, \quad (2)$$

where N represents amount of gas in a specific node, X_i is the mole fraction of the gas inside the node, $X_{i,in}$ is node inlet gas mole fraction, $X_{i,out}$ is node outlet gas mole fraction.

II Energy conservation model.

The temperature of positive-electrolyte-negative (PEN) is used as the variable in the solid layer model. According to the law of energy conservation, the temperature dynamic process of the solid control unit is expressed as follows^[24]:

$$\frac{dT_{PEN}}{dt} = (\rho_{PEN} V_{PEN} C_{PEN})^{-1} \cdot (\dot{Q}_{react} + \sum \dot{Q}_{in,PEN} - \dot{W}_{out}), \quad (3)$$

where $\dot{Q}_{react}$ and $\dot{W}_{out}$ represent the heat transfer and the energy exhaust for discharge in PEN, respectively. $\dot{Q}_{in,PEN}$ is the inlet energy of PEN; T_{PEN} is PEN temperature; ρ_{PEN} is PEN mass density; C_{PEN} and V_{PEN} are PEN specific heat capacity and volume, respectively. The new generated energy is from the energy and heat energy of the hydrogen with chemical energy. In the metal linker unit, new generated energy does not exist. The heat transfer between the solid control unit and the adjacent unit is divided into two ways. The first one is that heat transfer is caused by heat exchange between the solid layer and adjacent unit; the second is that the gas is a medium for heat transfer.

III Electrical characteristics model.

In general, the voltage of a stack often consists of three voltage values. These three items are ohmic voltage, activation voltage, and concentration voltage loss, respectively. Modeling based on this method has been described in some literatures^[20, 28, 30–31]. However, the expressions of stack model are complex, which is not conducive to modeling. Therefore, the equivalent resistance^[32] is used to model the electrical characteristics of the SOFC stack. The equivalent resistance R_{tot} is defined as follows:

$$R_{tot} = 0.0028 + 0.2796x - 0.9274x^2 + 1.5921x^3 - 0.8775x^4, \quad (4)$$

where $x = 0.001T_{PEN}/I_s$, I_s is the set current value and T_{PEN} is the PEN layer temperature.

The stack voltage U_{stack} expression is as follow:

$$U_{stack} = n \cdot (U_{OCV} - I_s R_{tot}) =$$

$$n \cdot (E^0 + \frac{RT_{PEN}}{2F} \ln(\frac{P_{H_2} P_{O_2}^{0.5}}{P_{H_2O}}) - I_s R_{tot}), \quad (5)$$

where F is Faraday constant, R is ideal gas constant, P represents partial pressure of gas, T_{PEN} is the PEN temperature, n is the number of cell, U_{OCV} is open circuit voltage of single cell.

3.2.1 Blower model and lagged model of gas supply subsystem

The main function of the blower is to blow air into the SOFC system. Different air flow rates are correspond to different blower power. The blower model is designed to subtract the parasitic losses from the overall efficiency of the system. The demanded blower power P_{blower} is expressed as follows^[9]:

$$P_{blower} = \frac{C_{p,air} \cdot T_{air}}{\mu \cdot (\dot{N}_{air})^{-1}} \left(\left(\frac{P_{blower.out}}{P_{blower.in}} \right)^{\frac{\alpha-1}{\alpha}} - 1 \right), \quad (6)$$

where $C_{p,air}$ is air isobaric heat capacity, T_{air} is air temperature, μ is blower effective, $P_{blower.in}$ and $P_{blower.out}$ are inlet and outlet air pressure, respectively, α is the ratio between air isobaric heat capacity and isometric specific heat capacity, $\dot{N}_{air}$ is the molar flow rate of the gas emitted from the blower. The lagging model mainly refers to the response characteristics of the system, when the gas flow through the flow meter and other components. For the simplification of the model, this paper uses a first-order inertia link plus delay. The hysteresis model is built^[31]:

$$G(s) = \frac{1}{Ts + 1} e^{-\tau s}, \quad (7)$$

where T represents a first-order inertia time constant, τ is a delay time constant.

3.2.2 Heat exchanger model

To exhaust gas manifold temperature, the main influence factor contain the heat conduction $\dot{Q}_{smoke, smokePipe}$ between exhaust gas and manifold and the heat conduction $\dot{Q}_{self, smokePipe}$ among the node of every manifold. The heat conduction equation is as follow^[33]:

$$\frac{dT_{smoke, smokePipe}}{dt} = (C_{smokePipe} \cdot m_{smokePipe})^{-1} \times (\dot{Q}_{smoke, smokePipe} + \dot{Q}_{self, smokePipe}), \quad (8)$$

where $C_{smokePipe}$ denotes exhaust gas manifold specific heat capacity, $m_{smokePipe}$ represents the node mass of exhaust gas manifold.

To exhaust gas temperature, the influence factor contains the heat conduction $\dot{Q}_{smokePipe, smoke}$ between exhaust gas manifold and exhaust gas, the heat conduction $\dot{Q}_{airPipe, smoke}$ between air manifold and

exhaust gas and the heat conduction among the material transmission^[33]:

$$\frac{dT_{\text{smoke}}}{dt} = (C_{\text{smoke}} \cdot N_{\text{smoke}})^{-1} \cdot (\dot{Q}_{\text{smoke, in}} + \dot{Q}_{\text{smokePipe, smoke}} + \dot{Q}_{\text{airPipe, smoke}} - \dot{Q}_{\text{smoke, out}}), \quad (9)$$

where C_{smoke} denotes the exhaust gas specific heat capacity, which can be calculated with Eq. (10); N_{smoke} is the mass of exhaust gas in the nodes.

The related gas specific heat capacity can be obtained by the following equation:

$$C_f = \sum X_i C_i(T) - R, \quad i \in \{\text{H}_2, \text{O}_2, \text{N}_2, \text{H}_2\text{O}\}, \quad (10)$$

where $C_i(T)$ represents specific heat capacity of gases at constant pressure, which can be seen in literature^[34]; C_f is the volume specific heat capacity; R is general gas constant ($8.314 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$).

To air manifold temperature, the influence factor contains exhaust gas convection heat $\dot{Q}_{\text{smoke, airPipe}}$, the manifold heat conduction $\dot{Q}_{\text{self, airPipe}}$ between the front and rear nodes and the heat conduction $\dot{Q}_{\text{air, airPipe}}$ between air and manifold^[33]:

$$\frac{dT_{\text{airPipe}}}{dt} = (\dot{Q}_{\text{smoke, airPipe}} + \dot{Q}_{\text{self, airPipe}} + \dot{Q}_{\text{air, airPipe}}) \cdot (C_{\text{airPipe}} \cdot m_{\text{airPipe}})^{-1}, \quad (11)$$

where C_{airPipe} denotes specific heat capacity of the air manifold, m_{airPipe} is the mass of air manifold nodes.

To air temperature, the influence factor contains the heat conduction $\dot{Q}_{\text{airPipe, air}}$ between air manifold and air and the introduction of heat is^[33]

$$C_{\text{air}} N_{\text{air}} \frac{dT_{\text{air}}}{dt} = \dot{Q}_{\text{airPipe, air}} + \dot{Q}_{\text{air, in}} - \dot{Q}_{\text{air, out}}, \quad (12)$$

where C_{air} is the air specific heat capacity, N_{air} is the mass of air in the nodes.

3.2.3 Burner model

The exhaust gas burner is mainly used for the recovery of exhaust gas in the system and to provide heat for the system. Electrochemical reactions occur in the stack interior. The remaining fuel is produced in the exhaust gas burner with air to produce high temperature exhaust gas, which is used to preheat the heat exchanger in pure hydrogen system and natural gas system. In this paper, the lumped parameter model of the exhaust gas burner is used. The burner model contains the temperature of the exhaust gas burner wall and the temperature of the exhaust gas.

The relation equation between temperature of ex-

haust gas T_{gas} and burner steel wall heat $\dot{Q}_{\text{wall}}$ is as follow^[33]:

$$N_{\text{gas}} \cdot \frac{dT_{\text{gas}}}{dt} = \dot{N}_{\text{in}} \sum X_{i, \text{in}} - \dot{N}_{\text{out}} \sum X_{i, \text{out}} + \frac{\dot{Q}_{\text{wall}}}{c_{\text{burner}}}, \quad (13)$$

where N_{gas} is the mass of exhaust gas in the burner, T_{gas} is the exhaust gas temperature in the burner, $\dot{N}_{\text{in}}$ and $\dot{N}_{\text{out}}$ are the mole flow of inlet and outlet burner gas, $X_{i, \text{in}}$ and $X_{i, \text{out}}$ are the mass of inlet and outlet burner gas, i is the gas type, c_{burner} is the average specific heat capacity in the burner.

The steel wall temperature T_{wall} has only the relation with the exhaust gas. The equation is described as follow^[33]:

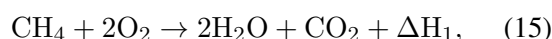
$$\dot{Q}_{\text{wall}} = \rho_{\text{wall}} C_{\text{wall}} \cdot \frac{dT_{\text{gas}}}{dt}, \quad (14)$$

where ρ_{wall} denotes the steel wall temperature, C_{wall} denotes the steel wall specific heat capacity.

3.2.4 Reformer model

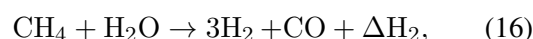
The heat supply of reformer is ignition. The main reaction in the process of heating and reforming is natural gas (the main gas is methane, the purity is 95%). The deionized water is evaporated into steam to reform.

The reforming reaction equation is^[35]



where $\Delta H_1 = -802.3 \text{ kJ/mol}$.

The specific combustion process reaction equation is^[35]



where $\Delta H_2 = 206 \text{ kJ/mol}$.

The reaction process needs to provide a high temperature environment, which is heated by the combustion chamber in the reformer. The mathematical model is as follows^[36]:

$$\frac{dP_c}{d\tau} = \frac{R_c T_c}{V} (G_{c, \text{in}} - G_{c, \text{out}}), \quad (17)$$

$$\frac{dG_{c, \text{out}}}{d\tau} = A_0 \left(\frac{P_{c, \text{in}} - P_{c, \text{out}}}{L} - U\sigma \right), \quad (18)$$

$$\frac{d(p_c c_{vc} T_c)}{dt} = \frac{1}{V} [(G_c c_p T_c)_{\text{in}} - (G_c c_p T_c)_{\text{out}}] - \alpha_c \frac{A}{V} (T_s - T_c) + \frac{\Delta H}{V}, \quad (19)$$

where A is contact area between air flow and separator, A_0 is channel area, P is pressure, G presents the flow rate, V is airflow channel volume, L is the length of reformer, T is the temperature, s presents the separator, c is the combustion, ΔH is variation of enthalpy, c_p is the specific heat at constant pres-

sure, U is the length of channel, the unit area frictional resistance coefficient is $\sigma = 1/2 f p v^2$, f is frictional drag coefficient, which can be approximated by $f = 16/Re$, $Re = p v L / \mu$, $G = p v A_0$.

The model of the reforming reaction is expressed as follows^[35]:

$$\frac{d(\rho_r y_j)}{d\tau} = \frac{(G_r y_j)_{in} - (G_r y_j)_{out}}{V} + \frac{W_j \cdot \text{rate}_j}{V}, \quad (20)$$

$$\frac{dG_{r,out}}{d\tau} = A_0 \left(\frac{P_{r,in} - P_{r,out}}{L} - U\sigma \right), \quad (21)$$

$$\frac{d(p_r c_s T_r)}{dt} = \frac{1}{V} [(G_r c_p T_r)_{in} - (G_r c_p T_r)_{out}] + \alpha_r \frac{A}{V} (T_s - T_r) + \frac{\Delta H \cdot \text{rate}_r}{V}, \quad (22)$$

where j represents the kind of gas, y_j is the mass fraction of j gas, W_j is the mole mass of gas j , rate is the chemical reaction rate, c_s is the set specific heat.

The separator mathematical model of reformer is described as follows^[35]:

$$\frac{dT_s}{d\tau} = \frac{\alpha_c A (T_c - T_s) - \alpha_r A (T_s - T_r)}{m_s c_s}, \quad (23)$$

where the heat transfer coefficient of the burner and the reformer chamber are α_c and α_r , respectively. The α_c and α_r can be calculated based on the flow characteristics of the fuel and air.

In addition, the reforming reaction is an endothermic reaction, so that H_2 and CO will absorb higher enthalpy. The reforming reaction is helpful to provide more heat enthalpy for the stack and after-burner.

3.2.5 SOFC system model validation

In order to ensure the correctness of the model, the SOFC system model is set to 1 kW system. The method of verification is to simulate the model dynamically and compare 1 kW SOFC experimental phenomena.

In the previous paper^[28], the initial parameters of the model are determined. In order to verify the validity and stability of the model, the integrated system model needs to be running. That is, in the normal input variables, the system can obtain a stable state during limited time. The selected input variable key parameters are shown in Table 2.

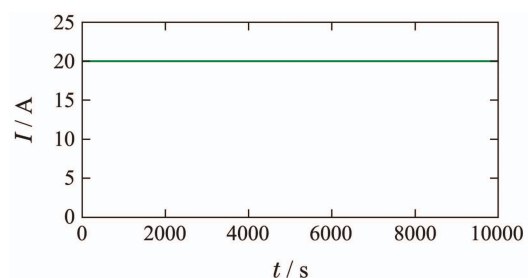
The initial N/O ratio is 0.79: 0.21, determined by the actual N/O ratio in air. The single cell voltage represents the low control limit of the cell, which is determined by the load power requirements and material property. According to the electrical characteristics of the stack, when the load required power is high, the stack operating voltage will be lower; when the load required power is low, the stack voltage will be high-

er. In order to obtain the stable dynamic process, the setting current is 20 A which is a constant.

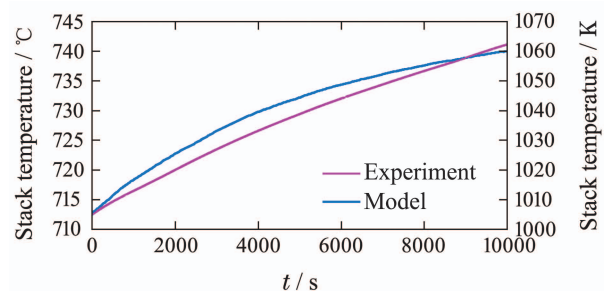
Table 2 Input variable key parameter

Parameter	Values
Fuel flow	0.02 mol/s
Fuel pressure	1.2×10^5 Pa
Air flow	1.04 mol/s
Air pressure	1.01×10^5 Pa
Air fuel temperature	293 K
The proportion of oxygen in the air	0.21
Single cell voltage	0.6 V
The proportion of nitrogen in the air	0.79

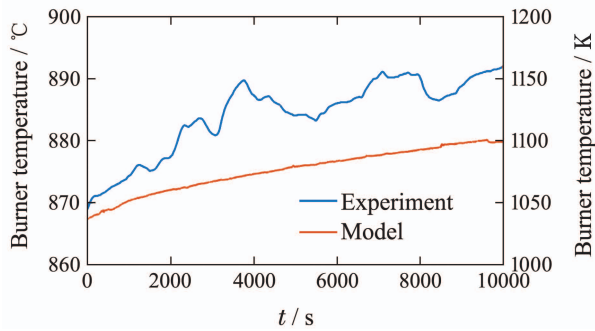
After determining the normal input variables, the system model key output including the stack current, stack temperature, burner temperature is determined. The system model simulation stop time is 10000 s, the model dynamic process and the actual experiment trend are shown in Fig. 3 and Fig. 4. In addition, the internal temperature of the stack is different along the direction of the airflow. The released heat from the electrochemical reaction will be brought out by the airflow. The outlet temperature of the stack can reflect effectively the temperature change of the stack, so the outlet temperature of the stack is chosen as the reference temperature of the SOFC stack. If each node temperature of the stack satisfies the temperature constraint condition, the stack temperature is satisfied constraint. And the simulation results are in line with the value of previous experimental experience, it can be considered that the model is verified. The detail trend is shown in Fig. 3 and Fig. 4. The blue line is experiment data, the unit is $^{\circ}\text{C}$, and the other colour line is model data line which unit is K .



(a) Discharge current

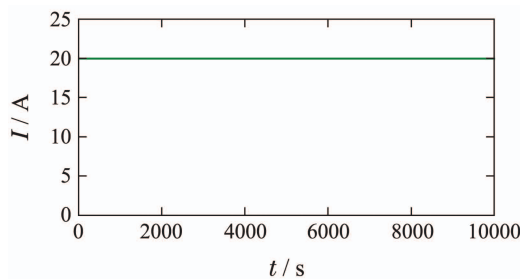


(b) Stack temperature

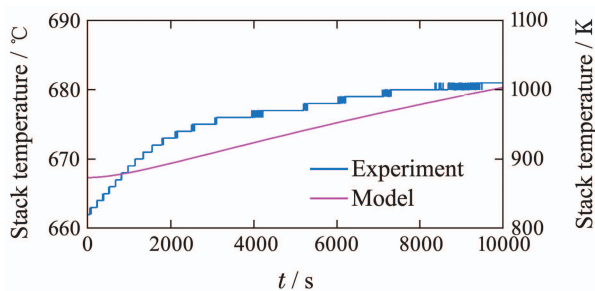


(c) Burner temperature

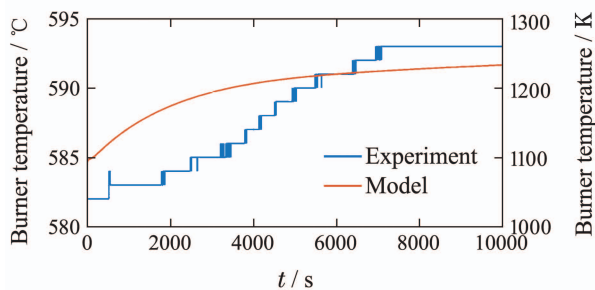
Fig. 3 Temperature of stack and burner in pure hydrogen system normal situation



(a) Discharge current



(b) Stack temperature



(c) Burner temperature

Fig. 4 Temperature of stack and burner in natural gas system normal situation

From Fig. 3 and Fig. 4, there are obvious differences between the experimental temperature and the model temperature. The reason for the temperature difference between the model and the actual system is that the system is ideal adiabatic, but the adiabatic effect of the actual system can not achieve absolute adiabatic. In Fig. 3 and Fig. 4, the experimental temperature is the average temperature data of six times experiment. The temperature of the model is the same

every time. Although the temperature difference between the stack and the burner is caused by thermal insulation, it can be seen from Fig. 3 and Fig. 4 that the trend of temperature change is the same, the difference is only reflected in the difference of temperature rising speed by thermal insulation. Therefore, we think that the established model is in line with the actual SOFC system.

3.3 Fault modeling

As the SOFC system consists of multiple components, there may be many faults, here we select the following two fault model to analyze: 1) the fuel volume blowing into the stack is reduction; 2) the air volume blowing into the stack is reduction. The analysis object is the stack temperature. The two selected the system structures are shown in Fig. 1 and Fig. 2, respectively. The main analysis is whether the stack temperature occurred abnormality.

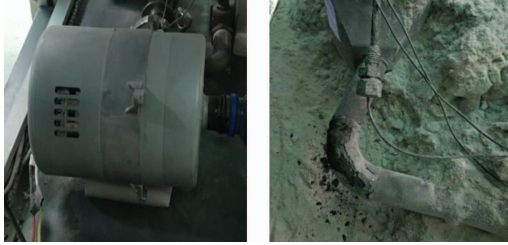
Fault 1 The fuel volume blowing into the stack anode is lower than ideal value.

There are two main reasons for this fault in two kind systems: the reduction in fuel pressure, resulting in a reduction in the amount of gas entering the stack per unit of time; manifolds are broken, transmission fault occurs, this fault is a strict actuator fault, the reason may be multifaceted, such as fuel bottle outlet gas leakage, heat exchanger manifold leakage, connecting the manifold leakage and so on. The hotbox of SOFC system operates in a high temperature environment. Although the gas manifold is not damaged by internal and external mechanical damage, the gas manifold uses metal materials. There still exists long-term high temperature work which leads to metal oxidation. After oxidation, the metal shell will be powdered, and eventually lead to metal rupture and fuel leakage. The manifold rupture caused by high temperature oxidation of the actual experiment is shown in the Fig. 5(a). So we select a manifold leakage fault between the heat exchanger outlet and the stack inlet in modeling. The model can be adjusted by reducing the air volume of blowing into the stack to 30%~95% of the nominal value to obtain the simulation results. For ease of analysis, the flow rate chosen to enter the stack is reduced to 90% of the normal value.

However, the two SOFC system structure dynamic respond in normal condition can be first seen in the Fig. 3 and Fig. 4 respectively. From Fig. 3 and Fig. 4, the temperature trend of hydrogen SOFC system is the same as natural gas system, and the burner temperature of the natural system is more stable than pure hydrogen system. The fault model is as follow^[37]:

$$F_{\text{fuel, act}} = F_{\text{fuel, normal}} \cdot (1 - \beta), \quad (24)$$

where $F_{\text{fuel, act}}$ denotes the blower real power after the fault occurs, $F_{\text{fuel, normal}}$ is the blower rating in normal situation, β is attenuation coefficient. We choose β to be 10%.



(a) Manifold leakage (b) Blower fault

Fig. 5 Actual SOFC system fault

Fault 2 The entering cathode air amount is lower than ideal value.

The amount of entering the stack cathode air is reduced, and the reason for the analysis may be multifaceted, such as air leakage or blower fault or the air mass flow controller fault. However, we find that the blower fault is the most common fault in actual experiment. In the process of blower running, the sealing is usually a key problem. It is well known that long-term operation of the blower can cause the bearing to gluing due to overheating. In order to avoid the occurrence of gluing, it is necessary to provide sufficient space for the blower to ensure heat exchange with external environment. External environment often has impurities and small particles, which will enter the blower due to the flow of air. These impurities will cause abnormal friction of blower bearings, resulting in bearing wear and eventually the bearing will be abnormal. With the wear increases, the running accuracy of the bearing is also continuously reduced. When the wear reaches a certain level, it will inevitably lead to significant decrease of air output at a certain power. The fault blower of SOFC system is shown in Fig. 5(b). So we choose the blower fault to analyze, which will result in the air amount reduction from blower under the original power. The adopted blower does not used as reactive power from SOFC stack. It is only necessary to multiply the wind blower expression by a certain coefficient in the modeling of the fault. The air flow rate at the outlet of the blower is reduced from the nominal value to 50%~95%. In order to facilitate analysis, we choose the blower of the air volume for the nominal value of 80%. The fault model is as follow^[37]:

$$P_{\text{blower, act}} = P_{\text{blower, normal}} \cdot (1 - \alpha), \quad (25)$$

where $P_{\text{blower, act}}$ denotes the blower real power after the fault occurs, $P_{\text{blower, normal}}$ is the blower rating in

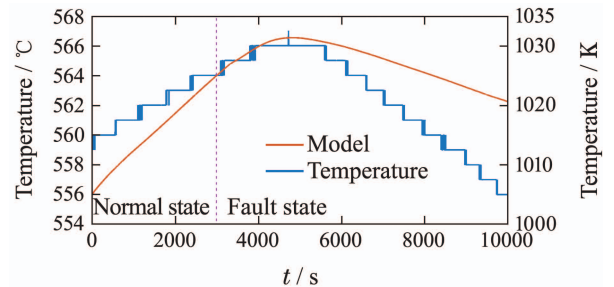
normal situation, α is attenuation coefficient. In order to simplify the simulation workload, selecting a constant as fault parameters to simulate here, that is we choose α as 20%.

4 Fault simulation and result

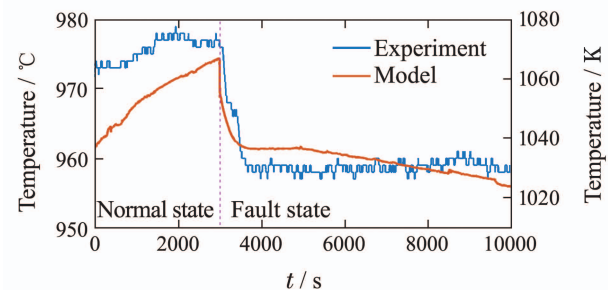
The overall models of the pure hydrogen and natural gas SOFC system are adjusted based on described model in Section 3.2. The adjust manner is that Eqs.(24)–(25) are added to the SOFC system model to simulate Fault 1 and Fault 2, respectively. In order to more effectively observe the fault model performance, the fault inserted time is set at 3000 s, the observe object is stack temperature and burner temperature. The set current is still 20 A. The select of other parameters are achieved in accordance with the expressions in Sections 3.3 and Table 2. The actual fault data is from the SOFC systems in HUST.

4.1 Simulation and result of Fault 1

The fuel flowing into the stack is set to normal, but fault occur at 3000 s, i.e. fuel supplied manifold leaks at 3000 s. The flow rate is decreased by 10%, and the pure hydrogen/natural gas temperature of the stack and the burner temperature is shown in Fig. 6 and Fig. 7, respectively.



(a) Stack temperature



(b) Burner temperature

Fig. 6 Temperature of pure hydrogen system stack and burner in Fault 1 situation

It can be seen from Fig. 6 and Fig. 7 that the temperature change trend of stack and burner in SOFC system model is consistent with actual system from the fault occurs beginning to end. In this way, the model of Fault 1 can be considered reasonable.

If other attenuation coefficient β is selected to simulate, both the stack temperature and the burner temperature will change. When β is larger, the two temperature values will be lower. When β is smaller, the temperature rises slightly compared with the current simulation result of β . But β does not affect the reference temperature change trend of SOFC system. For specific reasons, please see the discussion in the next section.

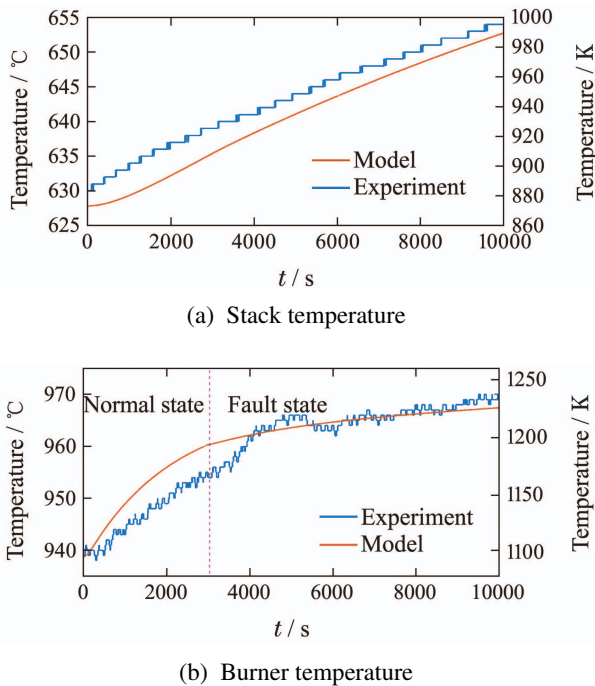


Fig. 7 Temperature of natural gas system stack and burner in Fault 1 situation

4.2 Simulation and result of Fault 2

The blower is set to normal, then the blower fault occur at 3000 s, the air flow rate decrease by 20%, and the pure hydrogen/natural gas temperature of the stack and the burner temperature is shown in Fig. 8 and Fig. 9, respectively.

It can be seen from Fig. 6 and Fig. 7 that the temperature change trend of stack and burner in SOFC system model is consistent with actual system from the fault occurs beginning to end. In this way, the model of Fault 2 can be considered reasonable.

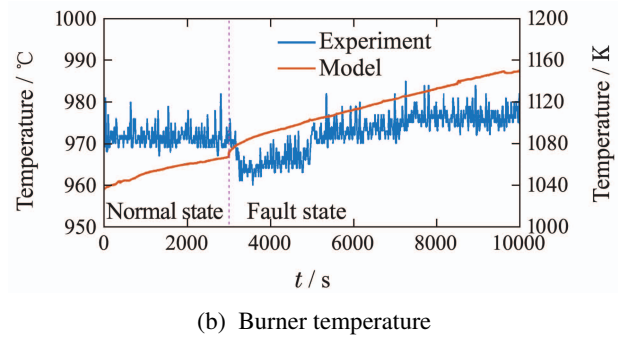
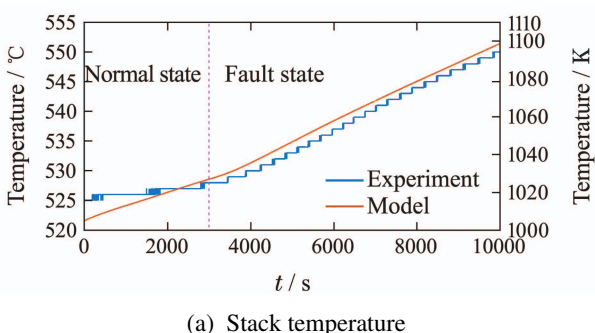


Fig. 8 Temperature of pure hydrogen system stack and burner in Fault 2 situation

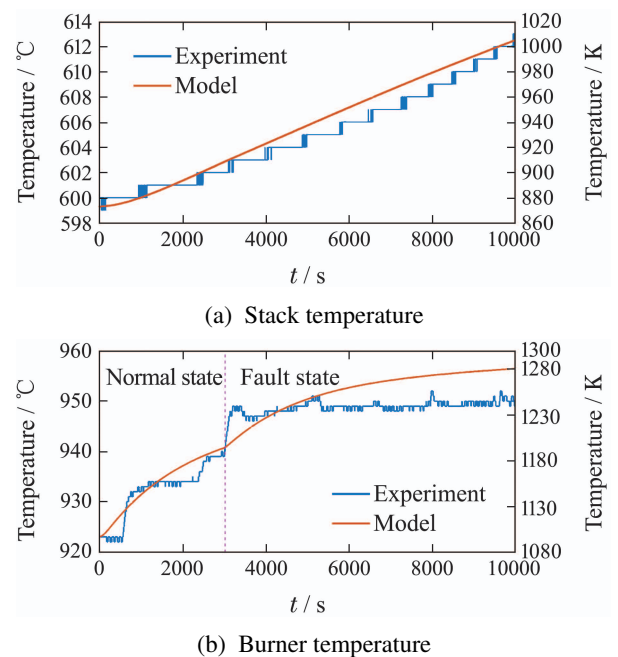


Fig. 9 Temperature of natural gas system stack and burner in Fault 2 situation

If other attenuation coefficient α is selected to simulate, both the stack temperature and the burner temperature will change. When α is larger, the two temperature values will be higher. When α is smaller, the temperature decrease slightly compared with the current simulation result of α . For specific reasons, please see the discussion in the next section.

5 Discussion

It can be seen from Figs. 6–9, when the stack current is constant at 20 A, the temperature curve of the stack and the burner are changed after fault is inserted at 3000 s. In addition, burner temperature strong fluctuation of natural gas SOFC system is caused by water evaporator, which is neglected here. The detailed fault analysis is as follows:

Fault 1 In pure hydrogen SOFC system, when the amount of fuel entering the stack is reduced suddenly, the temperature of the entire stack begins to show a downward trend at about 4200 s in Fig. 6(a).

The reason is that the available amount of fuel is decreasing in the stack, but the discharge is an exothermic reaction, it is helpful to keep the stack temperature. Therefore, the temperature drop of the stack is delayed. Less fuel enters the stack, the release of heat will naturally be reduced. When the heat is not enough to maintain the temperature of the stack, the temperature drop is reasonable. Since the thermal variation of the stack has a significant hysteresis, the stack temperature does not drop immediately when a fault occurs at 3000 s. But after a period of time the slope of the temperature change gradually decreases until the slope becomes negative, the stack temperature begin to decline. The model reflects the thermal characteristics of the stack well.

The temperature of the burner changes significantly in a short time. The main reason is that the burner has only a combustion effect, and there is no electrochemical reaction. So when the fuel enters the burner suddenly reduce, the temperature of the burner will suddenly appear a steep drop, and then maintain a certain slope decreasing gradually. The temperature change trend of burner is shown in Fig. 6(b). The model reflects the thermal characteristics of the burner well.

In natural gas SOFC system, the temperature dynamic characteristics of the stack and the burner are different to the pure hydrogen system from Fig. 7(a) and Fig. 7(b). The stack temperature of the natural gas system remain increases rapidly after the fault occurs, and the burner temperature is slow down and then continues to rise.

Due to the supplied natural gas decreases, it is helpful to improve the reformability of reformer. Furthermore, the reforming reaction is more sufficient, resulting in less methane and more hydrogen entering the stack, then more intense electrochemical reactions happen in the stack. As a result, the burner restored its fuel supply after a temporary reduction in fuel supply. Nevertheless, the amount of entering burner fuel is lower than before fault occurs. Therefore, the temperature of burner is rising, but the slope of temperature rise has been significantly reduced. The final temperature tends to be stable.

Fault 2 In pure hydrogen SOFC system, when the blower occurs fault, the amount of air entering into the stack will reduce. The temperature of the stack begins to a marked upward trend at about 3500 s, because there is not enough air to take away the heat in the stack and the exothermic reaction of the stack continues at the original rate. The air of blowing into the stack decreases, resulting in that the heat generat-

ed by the discharge can not be taken away. Since the thermal change of the stack has obvious hysteresis, the stack temperature does not rise immediately when the fault is inserted at 3000 s, but the slope of the temperature change gradually increases after a certain period of time. We can find from Fig. 8(a) that the stack temperature is higher than the normal state after the inserted blower fault occur at 3000 s. The experimental data and model show that the lag of temperature characteristics from blower fault is significantly lower than that caused by fuel manifold leakage.

From Fig. 8(b), the temperature of the burner has steep rise after a steep drop in a short time. The main reason is that the burner has only a combustion effect, and there is no electrochemical reaction, and combustion requires air support. So when the air blowing into the burner suddenly reduced, the burner temperature will suddenly appear steep drop. When the air flow rate is stable, the burner temperature will maintain a certain slope increased gradually.

In natural gas SOFC system, the temperature dynamic characteristics of the stack and the burner are different to the pure hydrogen system. The stack temperature trend of the natural gas system does not change obviously in Fig. 9(a), and the burner temperature tends to be stable after a steepened in Fig. 9(b). The reason is that a air volume temporary reduction causes heat to accumulate in burner.

The heat transfer capacity of heat exchanger is reduced due to the decrease of entering the system air amount. The heat generated by electrochemical reaction in SOFC stack can not enter the exhaust burner in time and quickly. The minimal reduction of air flow rate will not have a great impact on the reforming reaction. In addition, the reduction of air volume further results in that the heat generation by combustion can not be quickly removed from exhaust burner. More combustible gases remain in the burner, which causes the temperature of the burner to rise rapidly, and the rate of temperature rise is higher than that before the blower fault occurs.

Fault detection According to the above discussion, we can understand the phenomenon of SOFC system in the case of blower fault and fuel manifold leakage. According to the temperature characteristics of the stack and burner, it can be qualitatively judged whether the system fault is the above discussed fault. If more detailed data can be provided, it is also feasible to use this model to complete quantitative analysis. Moreover, due to the hotbox fault detect difficulty of SOFC system, combining the above mentioned attenuation coefficients α and β with the correct SOFC

model, is helpful to judge the fault grade of air blower or the leakage degree caused by the breakdown of fuel manifold. Therefore, it can be seen that the research is helpful to the fault detection of SOFC system in this paper. In addition, according to the comparison of Fig. 6(a) and Fig. 7(a), it can be found that the stack has a good fault-tolerant characteristic for fuel manifold leakage under natural gas system, and the performance of the stack will not change significantly.

6 Conclusions

In this paper, two kinds of fuel supplied SOFC system structure model is established and verified. The thermal characteristics of the two system are analyzed and the fault models are established for the fuel supply manifold leakage and blower fault, respectively. After running the 3000 s, the system fault is inserted, and the stack and burner thermal characteristics of the SOFC system are compared under different fuel supply modes. The results show that:

1) In the normal state, the stack temperature rise speed of the pure hydrogen system is not as good as that of the natural gas as fuel. Especially in the dynamic characteristics of the burner, the burner temperature of the natural gas SOFC system is also more stable, while the burner temperature of the pure hydrogen system has a slight temperature fluctuation.

2) Under the fuel manifold leakage fault, the stack temperature of the pure hydrogen system first rises and then drops, the burner has the characteristics of instant drop and gradual stable. While the stack temperature of the natural gas system keep the original heating rate after the fault occurs, and the temperature of burner remain increasing after instantaneous decrease.

3) Under the fault of the blower, the stack and the burner temperature rise of the pure hydrogen system increased obviously, while the stack temperature of the natural gas system did not change obviously, and the burner temperature appeared to be steepened and tended to be stable.

Therefore, it is concluded that the natural gas system is adaptive to the fault tolerance of the stack. In the future, this SOFC system fault model will be used to detect the system fault. And the model is also helpful to SOFC system type selection.

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